AtomSpect

Theory Manual

Zeeman and hyperfine spectra for plasma diagnostics

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This manual describes the physics and numerical methods implemented in AtomSpect: the within-term Zeeman Hamiltonian, hyperfine structure, axial dc Stark shifts, relative electric-dipole strengths, observation geometry, and continuous line profiles. It states the conventions of the inputs and outputs and cross-references the routines that implement each calculation. Numerical checks run with the code itself are in Section 9. Known limitations are collected in Section 10; input options are listed in Appendix A. The physical model is presented in the AtomSpect paper, doi:10.1016/j.jqsrt.2026.110015; implementation conventions and departures are identified here.

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1 Introduction

AtomSpect computes relative electric-dipole (E1) spectra for a selected lower and upper LS term in a static magnetic field, optionally including hyperfine structure and scalar/tensor dc Stark shifts along the quantization axis. The fields split and mix the supplied levels; every upper–lower eigenstate pair receives a wavelength, a relative strength, and a polarization label. Doppler and instrumental profiles turn the discrete ("stick") spectrum into a continuous curve.

The model uses four principal steps:

1. Field-free energies are supplied per J level, lowest J first; they are not calculated ab initio (Section 1.3).
2. Clebsch–Gordan coefficients transform the angular-momentum operators into the coupled basis (Sections 2 and 3).
3. Each isolated term is diagonalized separately, including the electronic and optional nuclear Zeeman interaction and supplied dc Stark terms.
4. E1 amplitudes between dressed states are weighted by the observation geometry, normalized, and broadened (Sections 5–7).

The ordinary spectrum uses angular strengths without radial matrix elements or level populations; collision dynamics are omitted. A separate calibrated API accepts external radiative data and explicit populations (Section 5.4). Ordinary strengths sum to one when the detected signal is nonzero; a dark observation returns zero strengths and a zero curve.

1.1 Read this before running

Development version operating status. The exact solver describes relative E1 spectra within isolated LS terms, with optional hyperfine structure and axial dc Stark terms. Absolute rates and explicit level populations are available through the separate atomic API, not ordinary `SpecOut`. Approximation spectra with HFS or nonzero electric field are unavailable. Finite convolution windows can lose broad wings and narrow peaks can be undersampled; check window and sampling convergence. See Section 10 for these restrictions and unvalidated research helpers.

The minimal example is `doc/scripts/example_minimal.py`. It computes He I 447 nm sticks at 0.5 T, reports their polarization labels, and convolves them with a split Lorentzian. Optional plotting is enabled with `-plot`. The returned `SpectrumResult` supports dictionary and attribute access; `result.describe()` documents the result keys.

1.2 Calculation flow

Figure 1 shows the operations performed by AtomSpect_Main; Table 1 maps them to the routines and sections that explain each stage.

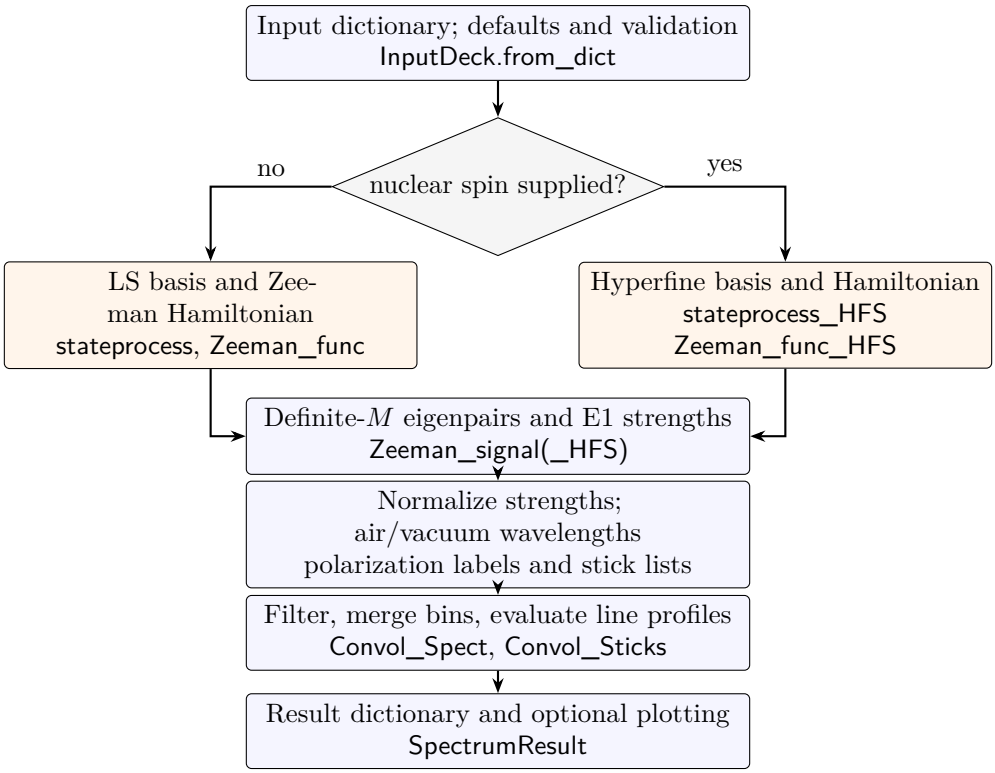


Figure 1: Calculation flow of AtomSpect_Main. Optional weak- and high-field spectra use separate non-HFS solves; energy fans sweep the Hamiltonian over field samples. Neither branch supplies absolute rates.

Step	Routines	Section
Defaults and input handling	InputDeck.from_dict	8
LS basis and Zeeman structure	stateprocess, Zeeman_func	2
Hyperfine structure	stateprocess_HFS, Zeeman_func_HFS	3
Angular E1 amplitudes	dipolestr, dipolestr_HFS	5
Observation geometry	polarization	6
Eigenpair strengths	Zeeman_signal, Zeeman_signal_HFS	5
Continuous profile	make_stickbins, Convol_Spect	7
Energy fans	ZeemanFan, FastZfan	2, 3
Numerical checks	`scripts/make_figures.py`, repository tests	9

Table 1: Where each part of the calculation is implemented and documented.

1.3 Units and conventions

Quantity	Unit	Note
Level energies, term energies	cm^{-1}	lowest J first
Magnetic field B	T	
Wavelengths	nm	air wavelengths by default; vacuum also returned
Hyperfine constants A, B	Hz	converted with $1/(100\,c)$
Temperature	K (`Temp`) or eV (`TempeV`)	$1\,\mathrm{eV} = 11602\ \mathrm{K}$
Line-profile functions	m	called with nm values scaled by 10^{-9}

Constants. $g_L = 1$, $g_S = 2.002319$, $\mu_\mathrm{B} = 0.466868\ \mathrm{cm}^{-1}\mathrm{T}^{-1}$ (from 5.7883818×10^{-5} eV/T and 1.23981×10^{-4} eV·cm). The nuclear g factor is $g_I = (\mu_I/I)(m_e/m_p)$ with μ_I in nuclear magnetons.

Spin input. The keys `s_ground` and `s_excited` are the spin quantum numbers S, not the multiplicities $2S + 1$: a triplet is entered as $S = 1$.

2 Zeeman structure

For a term with quantum numbers L, S, the code stores paired labels $[J, m_J, L, m_L, S, m_S]$, with $J = |L - S|, \ldots, L + S$, built from the Clebsch–Gordan coefficients $\langle Lm_L Sm_S | Jm_J\rangle$. These coefficients form the matrix Ψ that transforms between the coupled and uncoupled bases. The zero-field Hamiltonian is diagonal, $H_0 = \mathrm{diag}(E_J)$, with E_J repeated for the $2J + 1$ values of m_J.

2.1 Intermediate field (exact within the term)

$$H = H_0 + \mu_\mathrm{B} B\,(g_S S_z + g_L L_z), \qquad S_z \to \Psi\,\mathrm{diag}(m_S)\,\Psi^T, \quad L_z \to \Psi\,\mathrm{diag}(m_L)\,\Psi^T. \tag{2.1}$$

H is diagonalized numerically. The eigenvectors are expressed in the coupled basis. Because $S_z + L_z$ conserves m_J, m_J remains a good quantum number at every field, while J mixes.

Implementation. `stateprocess` constructs the basis transformation; `Zeeman_func` diagonalizes the Hamiltonian. `HZeeman` returns the unscaled Zeeman operator for repeated field solves. Multiply it by $\mu_\mathrm{B} B$ before adding it to the zero-field energy matrix.

2.2 Low-field approximation

When $\mu_\mathrm{B} B \ll$ the fine-structure splittings, $E = E_J + g_J m_J \mu_\mathrm{B} B$ with

$$g_J = g_L \frac{J(J+1) - S(S+1) + L(L+1)}{2J(J+1)} + g_S \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}, \qquad g_{J=0} = 0. \tag{2.2}$$

The electron g factors are the same as in the exact Hamiltonian, so the exact result approaches this one at weak field. Fan rows follow ascending energy rank at each sample; individual states are not tracked through level crossings.

2.3 High-field (Paschen–Back) approximation

When $\mu_B B$ is much larger than the fine-structure splittings, the levels are $E = E_{\text{term}} + \mu_B B\,(g_L m_L + g_S m_S)$, diagonal in the uncoupled basis; the eigenvectors are rotated back to the coupled basis with Ψ. This regime requires the term energy E_{term} (`EtermG`, `EtermE`) and neglects fine structure.

Caveat. The high-field approximation uses the ideal $g_S = 2$ of the uncoupled expression and ignores fine structure entirely. It is an asymptotic illustration, not a substitute for the exact intermediate-field result. In the driver it produces strengths and wavelengths only; its polarization labels are not resolved by eigenstate pair.

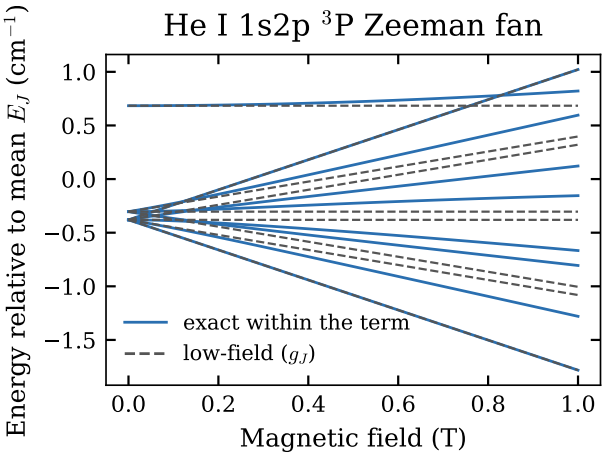


Figure 2: Zeeman fan of the He I 1s2p ^{3}P term (inputs from the repository example). The exact levels (solid) approach the g_J low-field lines (dashed) at weak field and mix J near 1 T.

3 Hyperfine structure

With nuclear spin I the bookkeeping lists are $[F, m_F, J, m_J, I, m_I, L, m_L, S, m_S]$, with bases related by two successive couplings, $(L, S) \to J$ and $(J, I) \to F$; the matrix Ψ is the product of the two Clebsch–Gordan matrices.

3.1 Zero-field hyperfine energies

With $K = F(F+1) - I(I+1) - J(J+1)$, the shift of a level of a given J is

$$\Delta E_{\text{hfs}} = \frac{AK}{2} + B\,\frac{\frac{3}{4}K(K+1) - I(I+1)J(J+1)}{2I(2I-1)J(2J-1)}, \tag{3.1}$$

and the quadrupole term is set to zero when $I \le \frac{1}{2}$, $J \le \frac{1}{2}$ or $B = 0$. A and B are given per J level in Hz and converted to cm^{-1}. This diagonal term is added to H_0 in the $|F, m_F\rangle$ basis.

3.2 Zeeman Hamiltonian with hyperfine structure

$$H = H_0 + H_{\text{hfs}} + \mu_B B\,(g_S S_z + g_L L_z - g_I I_z), \tag{3.2}$$

diagonalized exactly. Here the full Ψ transformation projects S_z, L_z and I_z into the coupled basis. The projected operators conserve m_F.

Implementation. stateprocess_HFS assigns the supplied A and B constants per J level. Zeeman_func_HFS includes their diagonal shifts and the electronic and nuclear Zeeman operators. The legacy `PsiHFS` input does not select a different Hamiltonian.

3.3 Weak-field hyperfine approximation (fans only)

Projecting the electronic and nuclear operators onto $|F, m_F\rangle$ gives

$$g_F = g_J \frac{F(F+1) - I(I+1) + J(J+1)}{2F(F+1)} - g_I \frac{F(F+1) + I(I+1) - J(J+1)}{2F(F+1)}, \qquad g_{F=0} = 0. \tag{3.3}$$

The shift is $g_F m_F \mu_B B$. The nuclear minus sign is shared with the exact Hamiltonian. A spin-zero nucleus has zero magnetic moment and $g_I = 0$.

3.4 Hyperfine Paschen–Back approximation (fans only)

For field sweeps, `ZeemanFan` can also return an approximation in which J is kept and I_z and J_z decouple: each diagonal element is the sum of the fine-structure energy, the A and B hyperfine terms evaluated for the uncoupled projections, and the electronic and nuclear Zeeman terms (with the nuclear sign above).

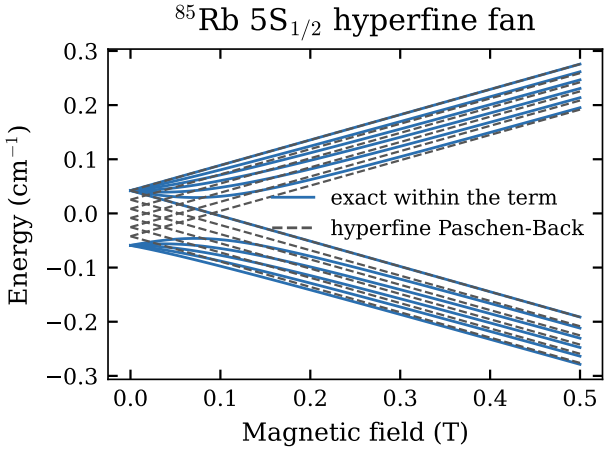


Figure 3: ^{85}Rb 5S$_{1/2}$ hyperfine fan. At $B = 0$ the two manifolds are separated by $A(I + \frac{1}{2})$; the approximation (dashed) is accurate once the Zeeman energy dominates the hyperfine splitting.

Caveat. Spectra with the low- or high-field approximation are not implemented with hyperfine structure; enabling `DoLowSig` or `DoHighSig` with `I_spin` raises `NotImplementedError`. The hyperfine constants and the nuclear moment are inputs and are not validated against tables by the code.

4 Axial dc Stark shifts

Following Steck's Rb-87 Eqs. (30)–(32) [4], take the electric field along the quantization axis, parallel to the magnetic field when present. With $a_0 = \alpha_0/h$ and $a_2 = \alpha_2/h$, the retained interaction is

$$\frac{H_{\mathrm{dc}}}{h} = -\frac{\mathcal{E}^2}{2}\left[a_0 + a_2 \frac{3J_z^2 - J(J+1)}{J(2J-1)}\right]. \tag{4.1}$$

Here angular momenta are dimensionless, $\mathcal{E}$ is in V/cm, and the supplied a_0, a_2 are in Hz/(V/cm)^2. The tensor term is zero for $J < 1$. These quantities are polarizabilities divided by Planck's constant, not polarizabilities in SI or atomic units; the symbol a_0 here is unrelated to the Bohr radius.

stark_hamiltonian forms the diagonal electronic $|J, m_J\rangle$ operator. With HFS it transforms this operator to $|F, m_F\rangle$ with Clebsch–Gordan coefficients. The full Hamiltonian $H_0 + H_{\mathrm{hfs}} + H_B + H_{\mathrm{dc}}$ is diagonalized before computing E1 amplitudes. The tensor term can mix F states with the same m_F; it is not merely an additive weak-field correction to the final eigenvalues. At zero electric field the existing magnetic-only path is used unchanged.

Implementation. `Efield` defaults to zero. At nonzero field, `alpha0_G` and `alpha0_E` are required; `alpha2_G` and `alpha2_E` default to zero. Each is a scalar or one value per electronic J, lowest J first. Supply absolute level polarizabilities: differential values from a table must first be added to the lower-state scalar polarizability. Tensor entries must vanish for $J < 1$. The exact spectrum and energy fans support these inputs; approximation flags must be disabled.

For a Rb-87 5S–5P deck, central Table 6 values give:

```
deck.update(Efield=1000.0, alpha0_G=0.0794,
            alpha0_E=[0.0794+0.122306, 0.0794+0.1340],
            alpha2_E=[0.0, -0.0406])
```

The remainder of the deck must already specify Rb-87 energies, mass and HFS constants. Cross-J Stark tensors and nonparallel electric/magnetic fields are not represented. The quadratic polarizabilities and retained electronic levels must remain appropriate at the applied field. Low-level Zeeman_func and Zeeman_func_HFS accept `Efield`, `alpha0` and `alpha2`; the calibrated API provides add_dc_stark with a level-ID mapping of (a_0, a_2).

5 Transition strengths

5.1 Basis-pair dipole factors

Without hyperfine structure the reduced dipole factor between a lower state $|J, m_J, L, \dots\rangle$ and an upper state $|J', m'_J, L', S, \dots\rangle$ (Metcalf, Eq. 4.32) is

$$d = (-1)^{L'+S-m'_J}\sqrt{(2J'+1)(2J+1)}\begin{pmatrix} J & 1 & J' \\ m_J & q & -m'_J \end{pmatrix}\begin{Bmatrix} L' & J' & S \\ J & L & 1 \end{Bmatrix}, \qquad q = m'_J - m_J. \tag{5.1}$$

Without hyperfine structure the code evaluates this expression with the roles of the two states exchanged; that changes only a sign that is common to all pairs of one q channel, and so drops out of the squared amplitude of a definite-M eigenpair. With hyperfine structure (Metcalf, Eq. 4.33) the factor contains two $6j$ symbols, one for the F–J coupling and one for the J–L coupling:

$$d = (-1)^{1+L'+S+J+J'+I-m'_F}\sqrt{(2F+1)(2F'+1)(2J+1)(2J'+1)} \times \begin{pmatrix} F & 1 & F' \\ m_F & q & -m'_F \end{pmatrix}\begin{Bmatrix} L' & J' & S \\ J & L & 1 \end{Bmatrix}\begin{Bmatrix} J' & F' & I \\ F & J & 1 \end{Bmatrix}. \tag{5.2}$$

The $3j$ and $6j$ symbols come from `sympy`. The code does not include the radial integral, so only ratios of strengths are meaningful.

5.2 Eigenpair amplitudes and normalization

Let c^u and c^l be the eigenvector matrices of the upper and lower level, and let $\mathcal{A} = c^{u*} \otimes c^l$ (a Kronecker product, lower index fastest). For every basis pair p the coefficient is $s_p = \epsilon_{q_p}\, d_p$, where ϵ_q is the polarization amplitude of Section 6. The strength of eigenpair k is

$$I_k = \Big|\sum_p \mathcal{A}_{pk}\, s_p\Big|^2, \qquad \hat{I}_k = \frac{I_k}{\sum_{k'} I_{k'}}, \tag{5.3}$$

so that the reported `signal` sums to one if the denominator is nonzero. When both transmissions vanish, the strengths remain zero instead of dividing by zero. The wavelength follows from the eigenvalues, $\lambda_{\mathrm{vac}} = 10^7/(E_u - E_l)$ nm, and the air wavelength from the standard vacuum-to-air relation.

5.3 Vacuum and air wavelengths

The air conversion uses the 1993 revised Edlén relation in Steck's Eq. (3) [4], evaluated at the vacuum wavelength. In Vac_to_air, `pressure_pa` is in Pa, `temperature_c` is in degrees Celsius, and `water_vapor_pressure_pa` is water-vapor partial pressure in Pa, not relative humidity. Defaults are dry air at 101325 Pa and 20 degrees Celsius. air_to_vac numerically inverts the same relation with five fixed-point iterations under laboratory conditions. The spectrum driver uses the defaults; custom atmospheric parameters are exposed by the helpers. Air temperature is separate from the emitter kinetic temperature `Temp`. The formula is empirical and is not validated at arbitrary wavelengths or near its poles.

5.4 Calibrated absolute E1 rates

The separate `atomic_data/atomic` API accepts external level and radiative information. RadiativeReference accepts A in s^{-1}, absorption oscillator strength f, gf, $\log_{10}(gf)$, reduced strength, or a lifetime in seconds with an explicit branching fraction. For a branch $u \to l$, $A_{ul} = b_{ul}/\tau_u$; the total inverse lifetime is the sum over all decay channels. Degeneracy factors are included when converting level data to reduced strength.

For magnetic-state dipoles $d_{ul,q}$ in SI units, the angle-integrated rate is

$$A_{ul} = \frac{\omega_{ul}^3}{3\pi\epsilon_0\hbar c^3} \sum_q |d_{ul,q}|^2. \tag{5.4}$$

calculate_components exports these rates as `A_s1`, using vacuum frequencies and a consistent SciPy constants set. directional_rates gives rates per solid angle; emission_weights accepts explicit incoherent populations and optional photon-energy weighting; synthesize_spectrum produces densities per nm on a vacuum-wavelength grid. A supplied `RadiativeData` attaches `AtomicComponents` to the ordinary result; it does not replace `SpecOut`. Model A accepts measured level-specific g_J and independent radiative branches, Model B uses complete strict-LS terms with one orbital norm, and Model C accepts explicit coherent E1 and magnetic tensors.

The standard electronic reduced amplitude in this API is $\sqrt{2J_l + 1}$ times Steck's lower-J-normalized dipole. For Rb D lines this is $\sqrt{2}$; the strength is twice the square of the tabulated amplitude. References supplied as lifetimes, A, or f handle this convention automatically. Natural linewidths are not automatically derived from decay sums: for a stable lower state, the ordinary-frequency natural FWHM is $1/(2\pi\tau_u)$. The Lorentzian/Voigt width is currently supplied by the caller.

Caveat. Quantities such as `A_Ein`, `f_osc` and `S_line` are not valid absolute radiative quantities: without a radial matrix element, a normalized angular strength cannot give a transition probability. They are therefore only returned on request (`RadiativePlaceholders`), together with a warning. Their $2J' + 1$ normalization is also tied to the basis-state list, not to field-mixed states.

Implementation. Zeeman_signal and Zeeman_signal_HFS form relative strengths and wavelengths. The internal definite-M solver resolves polarization labels for exact spectra. Returned basis-pair lists remain bookkeeping arrays; they are not labels of the dressed eigenstates.

6 Polarization

Following Isler (1997), with θ the angle between the magnetic field and the line of sight and γ the angle of an optional linear polarizer measured from the π axis,

$$e_\alpha = \sqrt{T_\alpha}\cos\gamma, \qquad e_\beta = \sqrt{T_\beta}\sin\gamma \qquad \text{(filter present)}, \tag{6.1}$$

$$e_\alpha = \sqrt{T_\alpha}, \qquad e_\beta = \sqrt{T_\beta} \qquad \text{(no filter)}, \tag{6.2}$$

$$\epsilon_0 = e_\alpha \sin\theta, \qquad \epsilon_{\pm1} = \sqrt{\tfrac{1}{2}(e_\alpha^2\cos^2\theta + e_\beta^2)}. \tag{6.3}$$

T_α, T_β (`TAlpha`, `TBeta`) describe transmission of other optics and default to one. A pair is labelled by $q = M_u - M_l$: $q = 0$ is π light, $q = \pm1$ is σ light, and any other value has no E1 component and is labelled 3. The public stick lists keep the emitted-light sign convention, so `sigmaplus_sticks` collects $q = -1$.

Labels belong to eigenstate pairs. M (m_J, or m_F with hyperfine structure) is a good quantum number, so each eigenvector has a definite M. The code rebuilds the Hamiltonian from the eigensystem, diagonalizes each M block separately, and labels every eigenpair with the exact M of its two eigenvectors. Labelling the basis pairs instead, which an earlier version did, assigns false π/σ labels once states mix. Equal-energy eigenvectors may be replaced by a reconstructed basis, so the identity of an individual coincident line is not preserved; only the summed strength of coincident lines has meaning.

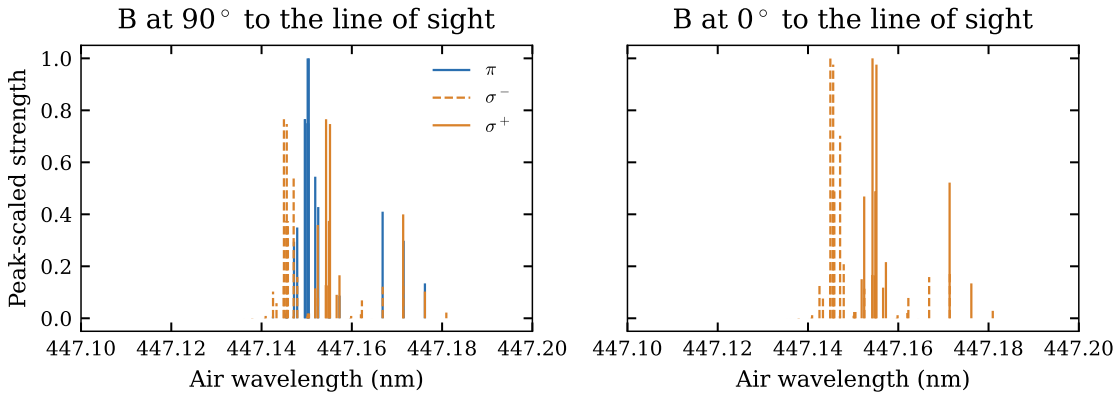


Figure 4: He I 447 nm sticks at 0.5 T split by polarization. Blue denotes π; orange denotes σ, solid for σ^+ and dashed for σ^-. With the field perpendicular to the line of sight all three kinds are visible; along the field the π light vanishes.

7 Continuous line profile

Sticks are convolved with unit-area profiles in metres. The instrument width `pixel_size` is a full width at half maximum; the input key `specstep` supplies it in nm. The pure Doppler Gaussian has no instrument-width argument, but uses `specstep` to set its sampling scale.

Name	Definition
`Gaussian`	Doppler: $\sigma = \lambda_0\sqrt{k_BT/(Mc^2)}$, mean $\lambda_0(1+v/c)$.
`GaussianInstrum`	Gaussian with FWHM $W = \sqrt{\mathrm{pix}^2 + 8\ln 2\,\sigma^2}$ (instrument and Doppler in quadrature).
`Voigt`	Doppler σ convolved with a Lorentzian of FWHM `pixel_size` (HWHM = pix/2 for `scipy`).
`Lorrentzian`, `Skewed`	Symmetric Lorentzian, or split Lorentzian with half widths $s_{L,R}\,\mathrm{pix}/2$ for `Skewed`; unit area.

Ion motion shifts the mean by $\lambda_0 v/c$; with `bidir` the profile is the average of the $\pm v$ components. All profiles integrate to one, so the total strength of a stick does not depend on the profile choice (Fig. 5).

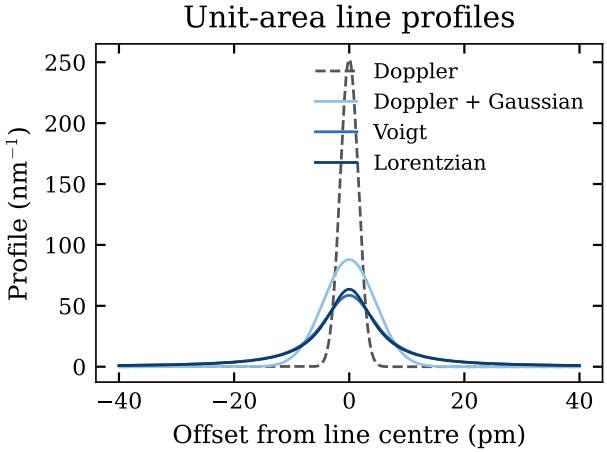


Figure 5: Unit-area profiles at 600 nm for a 3000 K, mass-40 emitter and a 10 pm instrument width.

Binning and truncation. For speed, Doppler-shifted centres separated by at most 2 `fxnwindow` (nm) share a bin, merging overlapping padded windows. Both counterstream centres participate in binning; a line is evaluated once at any point, even when its two streams share a bin. Each bin is evaluated within `fxnwindow` of its outermost shifted centres, at approximately `specres` points per pixel. All positive strengths are retained by default. Optional `relative_cutoff` drops components at or below that fraction of the largest input strength, without renormalization. The grid includes the bin endpoints and uses at least two samples; its spacing is approximately `specstep`/`specres`. Explicit zero samples immediately outside bin boundaries prevent interpolation and trapezoidal integration from creating artificial intensity across gaps. Profiles are summed in bounded chunks rather than one arbitrarily large matrix.

The Lorentzian tail beyond $\pm w$ carries the fraction $1 - \frac{2}{\pi}\arctan(w/\gamma)$ of a symmetric line's area (γ the HWHM); the input check warns when this exceeds 2%. For Voigt this reports only the Lorentzian-tail estimate, not the total finite-window loss. Binning remains a finite-support approximation: omitted tails are not redistributed or renormalized, and increasing the padding is needed to assess their contribution.

Density units. `SpecOut` has a wavelength axis in nm but retains profile values per metre. Its integral against the nm axis is therefore 10^9 times its integral against metres. For a density per nm, multiply the vertical values by 10^{-9}. Sampling, filtering and finite windows can reduce the captured area; the plotted curve is not renormalized.

Caveat. Binning follows shifted centres, but the `spec_window` filter and displayed sticks use rest wavelengths. Include all contributing rest centres in that selection window, including lines whose observed peaks move into the plot window. Gaussian tails can be truncated without a warning; a narrow thermal Gaussian may require more samples than the nominal instrument-based spacing. The first-order Doppler shift assumes $|v| \ll c$. These are optically thin sums of supplied line weights; collision dynamics, transfer, natural-width inference and detector pixel integration are not performed by the convolution routine.

Implementation. make_stickbins merges overlapping padded intervals; Convol_Sticks evaluates a unit-area profile within each interval; Convol_Spect weights and sums the curves. The wavelength axis is in nm while profile density remains per metre.

8 Using AtomSpect

Input. `AtomSpect_Main` takes a dictionary. The required keys are `s_ground`, `s_excited`, `l_ground`, `l_excited`, `E_ground`, `E_excited`, `Bmag` and `b_angle`. Unknown keys raise a warning with a suggestion. Level energies must match the allowed J values, lowest J first; HFS constants may be scalars or one per J. Widths, masses and sampling factors must be positive, and optical transmissions nonnegative. A pure Gaussian requires positive temperature. The Gaussian, `GaussianInstrum` and Voigt profiles need `Temp` or `TempeV`. The full list with defaults is in `AtomSpect_Main`'s docstring and in the README.

Output. The result is a dictionary with attribute access. Its type is SpectrumResult; call `result.describe()` for descriptions of its entries. Principal entries include `signal` and `wave_air` for the sticks, `rad` for polarization labels, and `SpecOut` for the continuous curve. Separate `pi`/`sigma` stick lists are also returned. `ZeemanSplits_G` is now a vector of lower-level energies in cm^{-1}; earlier code accidentally returned the eigenvector matrix. Exact basis metadata is retained when approximation spectra are also requested.

Module	Contents
`constants`, `units`	constants; energy and wavelength conversions
`angular`	g factors, Clebsch–Gordan and $3j/6j$ wrappers, dipole factors
`levels`	state lists and coupling matrices
`hamiltonian`, `hfs`	Zeeman and hyperfine Hamiltonians, eigenvector resolution
`stark`	axial scalar/tensor dc Stark matrices
`atomic_data`, `atomic`	radiative references; calibrated components and spectra
`polarization`, `transitions`	polarization amplitudes and labels; strengths and wavelengths
`fan`	energy fans versus B
`lineshapes`, `convolution`	profiles; binning and convolution
`config`, `spectrum`, `result`	input validation; the driver; the result class
`plotting`, `io`	plots and the slider GUI; reading and saving data

Table 2: Module map of the AtomSpect package.

9 Numerical validation

9.1 Analytic checks

The script `scripts/make_figures.py` evaluates the checks in Table 3 before writing any figure and aborts if one fails. They test specific limits of the implemented model; they do not establish the correctness of the whole code or its agreement with experiment.

Check	measured	tolerance
`weak_field_zeeman_3P2_cm-1`	8.73e-11	1e-09
`rb85_5S_hyperfine_interval_relative`	1.37e-16	1e-09
`strength_normalization`	0.00e+00	1e-12
`no_forbidden_components`	0.00e+00	1e-14
`pi_strength_along_field`	0.00e+00	1e-12
`gaussian_unit_area`	0.00e+00	1e-03
`voigt_unit_area`	7.96e-04	1e-03
`lorentzian_fwhm_m`	7.37e-23	2e-14
`hfs_weak_field_projection`	4.44e-16	1e-12
`overlapping_bin_profile_sum`	3.81e-16	1e-12
`dark_spectrum_strength`	0.00e+00	0e+00

Table 3: Analytic checks: weak-field splitting of $^3\mathrm{P}_2$ against $g_J m \mu_\mathrm{B} B$ (absolute, cm^{-1}); ^{85}Rb 5S hyperfine interval against $A(I + \frac{1}{2})$ (relative); normalization; absence of $|\Delta M| > 1$ strength; vanishing π light along the field; profile areas and widths; weak-field HFS projection; overlapping-bin profile summation; a dark observation.

The repository's test suite adds regression tests for the polarization labels (with and without hyperfine structure), the high-field and fan approximations, the Boolean approximation flags, the profiles and a set of golden outputs for refactoring.

`test_units.py` and `test_steck_rb87.py` additionally compare air indices, Rb-87 hyperfine intervals, Breit–Rabi energies, branch factors, representative dipole components, lifetime/oscillator conversions, and absolute radiation patterns. `test_stark.py` checks scalar field-squared shifts, tensor eigenvalues and trace, weak-HFS tensor expectations, an independent spin Hamiltonian through strong HFS mixing, basis permutations, fan/driver consistency, the radiative bridge, and exact equality with the existing output when the electric field is omitted or zero. These cases do not certify every signed table entry or establish experimental accuracy.

10 Known limitations and implementation notes

- Ordinary `SpecOut` uses relative angular strengths. Absolute rates, explicit populations and photon-energy weighting require the separate calibrated atomic API and supplied atomic data. Legacy radiative placeholders remain invalid absolute quantities.
- Each term is isolated. Mixing between terms of different L or S and configuration interaction are not modelled.
- Hyperfine constants and moments are user inputs. The estimator module `hfs_estimate` (semi-empirical A and B constants) is a research aid. Its B-constant routines evaluate $B = eQV_{zz}$ in SI units; the underlying hydrogenic A-constant model is pinned by tests but not validated.
- The non-HFS high-field approximation neglects fine structure and uses $g_S = 2$.

- `sortE` changes only the order in which basis states are listed, not the eigenvalues, and is slow.
- Without `Convfxn` the symmetric Lorentzian is used and `Temp` is ignored. `Skewness` is used only by `Skewed`.
- Air/vacuum conversion numerically inverts the revised Edlén relation, with validity limited to its empirical domain. The driver uses dry air at 20 degrees Celsius.
- Dc Stark shifts retain scalar and tensor polarizabilities diagonal in electronic J, for parallel fields only. They require supplied data. No cross-J Stark tensors, diamagnetism or collision dynamics are included.
- HFS includes A/B, but not octupole C or cross-J HFS tensors. Octupole shifts require both $I, J \geq 3/2$; no nonzero Rb-87 C is supplied by the cited Steck reference.
- Vapor pressure, recoil shifts, population kinetics and radiative transfer are not calculated. Density and optical depth matter for absolute intensity and self-absorption; recoil matters at precision/ultracold scales. The code does not infer these corrections from normalized sticks.
- Legacy magnetic constants and strict-LS g_J limit precision. The newer API can accept measured g_J or explicit magnetic operators. Partial decay sums represent inverse lifetimes only when all relevant channels are retained.
- `Calc_HFS_Vals` is a reserved input and does not estimate missing constants.
- The legacy `Zsig_norad` helper retains raw eigenvectors; individual values can depend on the basis chosen in degenerate subspaces. Use the labelled spectrum for polarization-resolved results.

10.1 Reproducing this manual

From the repository root, run:

```
python doc/scripts/example_minimal.py
python -m pytest -q
python doc/scripts/make_figures.py --data-only
python doc/scripts/plot_figures.py
python doc/scripts/build_manual.py
```

The data generator stages output until all checks and calculations succeed. The plotting script redraws all four PDFs from the CSV files. Provenance records UTC time, runtime versions, source commit, a source-content hash, and figure settings. Add `-rebuild` to the manual builder to regenerate data and figures before compiling. The Makefile offers equivalent data, plots, figures, check and rebuild targets. LaTeX failures stop the build.

A Options and physical parameters

Required entries are marked with a dash. Both terms use the same unit and ordering conventions. Full result-key descriptions are available from `result.describe()`.

Input	Default	Meaning (symbol, section)
`s_ground/s_excited`	—	spin S, not $2S+1$ (§1.3)
`l_ground/l_excited`	—	orbital angular momentum L (§2)
`E_ground/E_excited`	—	one energy per J in cm^{-1}, lowest J first
`Bmag`	—	scalar field [T]; field array for fans
`Efield`	0	axial dc electric field [V/cm], parallel to B (§4)
`alpha0_G/alpha0_E`	required if $\mathcal{E} \neq 0$	absolute scalar α_0/h $[\mathrm{Hz}/(\mathrm{V/cm})^2]$, scalar or one per J
`alpha2_G/alpha2_E`	0	tensor α_2/h in the same units; zero for $J < 1$
`b_angle`	—	angle between field and line of sight [degrees] (§6)
`Pol_angle`	absent	linear analyzer angle [degrees]; zero enables the filter
`TAlpha/TBeta`	1 / 1	nonnegative intensity transmissions
`amu`	1	emitter mass number
`Convfxn`	`Lorrentzian`	Gaussian, GaussianInstrum, Voigt, Skewed, or symmetric Lorentzian (§7)
`Temp/TempeV`	profile-dependent	temperature [K]/[eV]; eV takes precedence; required for thermal profiles
`specstep`	0.002222 nm	instrument FWHM and sampling pixel width; 0.002 nm if `Convfxn` is supplied
`specres`	10	nominal samples per pixel
`fxnwindow`	0.1 nm	padding of Doppler-shifted component bins (§7)
`relative_cutoff`	0	optional cutoff as a fraction of maximum input strength; no renormalization
`Skewness`	[1,1]	left/right half-width multipliers, only for Skewed
`ion_vel`	0	velocity [m/s]; scalar or counterstreaming pair
`bidir`	False	average $+v$ and $-v$, or $+v_1$ and $-v_2$
`SpectrumData`	absent	measured [wavelengths in nm, intensities]
`spec_window/plot_window`	[1,1000] nm	physics/plot bounds; default to measured wavelength range when supplied
`I_spin`	absent	nuclear spin I; enables HFS (§3)
`mu_I`	required with HFS	nuclear magnetic moment [nuclear magnetons]
`HFS_G/HFS_E`	required with HFS	dipole A constants [Hz], scalar or one per J
`HFSB_G/HFSB_E`	0	quadrupole B constants [Hz], scalar or one per J
`PsiHFS`	False	legacy setting; same Hamiltonian for both values
`Calc_HFS_Vals`	ignored	reserved; no constant estimation
`DoLowSig/DoHighSig`	False / False	non-HFS approximation spectra; Boolean or yes/no values
`EtermG/EtermE`	absent	term energies $[\mathrm{cm}^{-1}]$ for non-HFS high-field solves; zero is valid
`sortE`	False	order approximation bases by weak-field energy
`RadiativePlaceholders`	False	opt-in unvalidated legacy radiative outputs (§10)
`RadiativeData`	absent	external radiative data; attaches calibrated `AtomicComponents` (§5.4)

Input	Default	Meaning (symbol, section)
`plottitle`	absent	plotting title

References

[1] R. C. Isler, Phys. Plasmas **4**, 3 (1997), doi:10.1063/1.872095 (polarization of Zeeman components).

[2] H. J. Metcalf and P. van der Straten, *Laser Cooling and Trapping*, Eqs. 4.32–4.33 (dipole factors).

[3] D. A. Steck, *Rubidium 85 D Line Data*, `https://steck.us/alkalidata/` (hyperfine constants and structure).

[4] D. A. Steck, *Rubidium 87 D Line Data*, revision 2.3.4 (8 August 2025), `https://steck.us/alkalidata/rubidium87numbers.pdf` (air index, dc Stark interaction and radiative benchmarks).

[5] N. J. Stone, At. Data Nucl. Data Tables **90**, 75 (2005), doi:10.1016/j.adt.2005.04.001 (nuclear moments).

[6] H. Kopfermann, *Nuclear Moments* (hyperfine interaction).

[7] L. Nofs *et al.*, J. Quant. Spectrosc. Radiat. Transf. (2026), doi:10.1016/j.jqsrt.2026.110015 (the AtomSpect paper).